your success = my concern

Tuks.t: 1~2 testing

Pure Mathematics Paper 1: [Time 3hrs]

INSTRUCTIONS TO CANDIDATES

Write your name and surname on the answer paper.

A blue or black pen is required for all the writing.

You may use a pencil for any diagrams or graphs.

Answer all questions.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of the marks for this paper is 120.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

[Turn over]

Answer **all** questions

1a) If the roots of the equation $2x^2 - 7x + 4 = 0$ are α and β, find the values of $\frac{1}{\alpha} + \frac{1}{\beta}$ and $\frac{1}{\alpha\beta}$. [4]

b) Hence write down the equation whose roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ in the form $ax^2 + bx + c = 0$, where a, b and c are integers. [2]

2a) Evaluate $\log_3 7 \times \log_7 2 \times \log_2 3$. [3]

b) Solve, for x and y, the simultaneous equations

$$125^x = 25(5^y),$$

$$7^x \div 49^y = 1.$$ [6]

3. Solve the equation

$$2\ln(5 - e^{-2x}) = 1,$$

giving your answer correct to 3 significant figures. [4]

4. Solve the inequality

$$\frac{2x^2 - x - 19}{x^2 + 3x + 2} > 1.$$ [5]

5. The function f is defined by $\mathrm{f} : x \to 2x^2 - 12x + 13$ for $0 \le x \le A$, where A is a constant.

a) Express $\mathrm{f}(x)$ in the form $a(x + b)^2 + c$, where a, b and c are constants. [3]

b) State the value of A for which the graph of $y = \mathrm{f}(x)$ has a line of symmetry. [1]

c) When A has this value, find the range of f. [2]

The function g is defined by $\mathrm{g} : x \to 2x^2 - 12x + 13$ for $x \ge 4$.

d) Explain why g has an inverse. [1]

e) Obtain an expression, in terms of x, for $g^{-1}(x)$, stating its domain. [3]

6. The cubic polynomial $y = f(x)$ is such that the coefficient of x^3 is -1 and the roots of $f(x) = 0$ are -3, -1 and 1 whilst the value of the y-intercept is 2.

a) Sketch the curve $y = f(x)$ showing all the intersection with the axes. [2]

b) Sketch the following graphs showing clearly the coordinates of the intersection with the axis

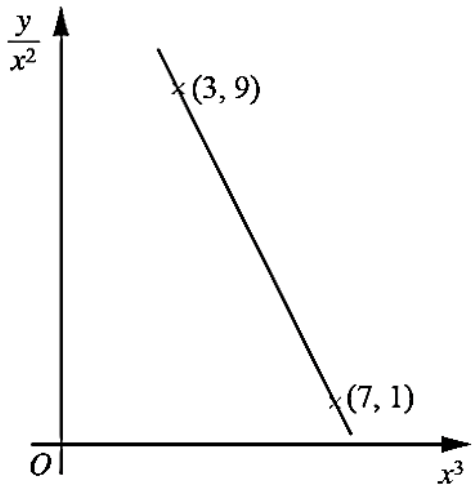
(i) $y = 2f(x)$, [2]

(ii) $y = f(x - 3)$. [2]

7a) Calculate the term independent of x in the binomial expansion of $\left(x - \frac{1}{2x^5}\right)^{18}$. [3]

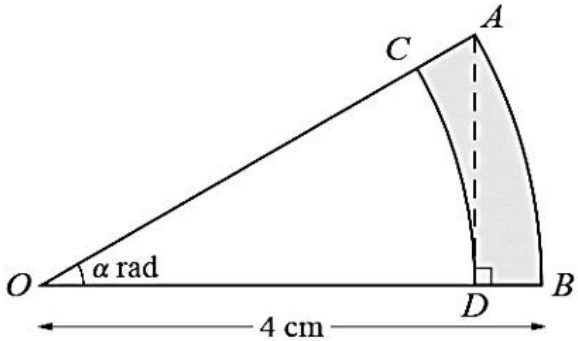
b) In the binomial expansion of $(1 + kx)^n$, where $n \geq 3$ and k is a constant, the coefficient of x^2 and x^3 are equal. Express k in terms of n. [4]

8.



The variables x and y are related so that, when $\frac{y}{x^2}$ is plotted against x^3, a straight line graph passing through $(3; 9)$ and $(7; 1)$ is obtained. Express y in terms of x. [4]

9.



In the diagram, AB is an arc of a circle with center O and radius 4 cm. Angle AOB is α radians. The point D on OB is such that AD is perpendicular to OB. The arc DC, with center O, meets OA at C.

a) Find an expression in terms of α for the perimeter of the shaded region $ABDC$. [4]

b) For the case where $\alpha = \frac{1}{6}\pi$, find the area of the shaded region $ABDC$, giving your answer in the form $k\pi$, where k is a constant to be determined. [4]

10. The complex number u is defined by $u = \frac{5}{a+2i}$, where the constant a is real.

a) Express u in standard form. [2]

b) Find the value of a for which $\arg(u^*) = \frac{3}{4}\pi$, where u^* denotes the complex congugate of u. [3]

11. If $y = \tan x$ find an approximation for δy when x is increased by δx.

Hence estimate, in terms of π, the value of $\tan\frac{9}{32}\pi$. [4]

12. Express $\frac{5x^3-3x^2+7x-3}{(x^2+1)^2}$ in the form $\frac{Ax+B}{x^2+1} + \frac{Cx+D}{(x^2+1)^2}$ where A, B, C and D are constants. [3]

13. A television quiz show takes place every day. On day 1 the prize money is \$1000. If this is not won the prize money is increased for day 2. The prize money is increased in a similar way every day until it is won. The television considered the following two different models for increasing the prize money.

Model 1: Increase the prize money by \$1000 each day.
Model 2: Increase the prize money by 10% each day.

On each day that the prize money is not won the television company makes a donation to charity. The amount donated is 5% of the value of the prize on that day, After 40 days the prize money has still not been won. Calculate the total amount donated to charity

(i) if Model 1 is used, [4]

(ii) if Model 2 is used. [3]

14a) Given that $15cos^2\theta + 2sin^2\theta = 7$, show that $tan^2\theta = \frac{8}{5}$. [4]

b) (i) Express $4\cos\theta + 3\sin\theta$ in the form $R\sin(\theta + \alpha)$, where $R > 0$ and $0° < \alpha < 90°$, giving the value of α correct to 2 decimal places. [3]

(ii) Hence solve the equation

$$4\cos\theta + 3\sin\theta = 4.5\ ,$$

giving all solutions in the interval $0° \leq \theta \leq 360°$, correct to 1 decimal place. [4]

(iii) Write down the least value of $4\cos\theta + 3\sin\theta + 7$ as θ varies. [1]

15. The equation of a curve is $\ln(xy) - y^3 = 1$.

a) Show that $\frac{dy}{dx} = \frac{y}{x(3y^3-1)}$. [4]

b) Find the coordinates of the point where the tangent to the curve is parallel to the y-axis, giving each coordinate correct to 3 significant figures. [4]

16. The equation of a curve is $y = x^2 - 4x + 7$ and the equation of a line is $y + 3x = 9$. The curve and the line intersect at the points A and B.

a) The mid-point of AB is M. Show that the coordinates of M are $\left(\frac{1}{2}; 7\frac{1}{2}\right)$. [4]

b) Find the coordinates of the point Q on the curve at which the tangent is parallel to the line $y + 3x = 9$. [3]

c) Find the distance MQ. [1]

17a) (i) Show that

$$\int_0^{\frac{1}{4}\pi} \cos 2x \ dx = \frac{1}{2}.$$ [2]

(ii) Find the exact value of

$$\int_0^{\frac{1}{4}\pi} \sin^2 x \ dx$$ [3]

b) (i) Use the trapezium rule with 2 strips to estimate the value of

$$\int_0^{\frac{1}{4}\pi} \sec x \ dx,$$

giving your answer correct to 2 significant figures. [3]

(ii) Determine, by sketching the appropriate part of the graph $y = \sec x$, whether the trapezium rule gives an under-estimate or an over-estimate of the true value. [2]

18. The curve $y = \frac{\sin x}{x}$ in the range $0 < x \leq 2\pi$ has its minimum point M.

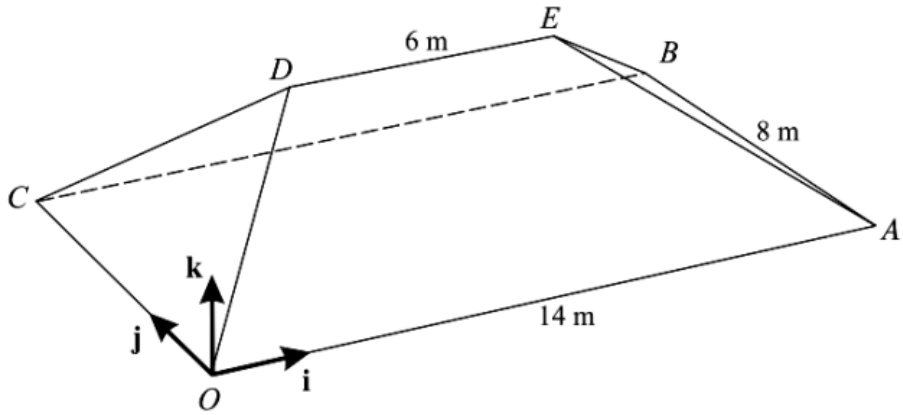
a) Show that the x-coordinate of M satisfies the equation $x = \tan x$. [4]

b) The iterative formula

$$x_{n+1} = \tan^{-1}(x_n) + \pi$$

can be used to determine the x-coordinate of M. Use this formula to determine the x-coordinate of M correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

19.



The diagram shows the roof of a house. The base of the roof, $OABC$, is rectangular and horizontal with $OA = CB = 14$ m and $OC = AB = 8$ m. The top of the roof DE is 5 m above the base and $DE = 6$ m. The sloping edges OD, CD, AE and BE are all equal in length.

Unit vectors **i** and **j** are parallel to OA and OC respectively and the unit vector **k** is vertically upwards.

a) Find the unit vector in the direction of $\overrightarrow{OD}$ in terms of **i**, **j** and **k**. [4]

b) Find the angle DOB correct to the nearest 0,1°. [4]

End of test